

NDVI & EVI Vegetation Analysis - Ibarapa East LGA, Oyo State, Nigeria

Project Overview

This project applies remote sensing and raster analysis in QGIS to examine vegetation characteristics within Ibarapa East Local Government Area, Oyo State, Nigeria.

Two commonly used vegetation indices were calculated from raster satellite data:

- Normalized Difference Vegetation Index (NDVI)
- Enhanced Vegetation Index (EVI)

The resulting raster layers were visualized to examine the spatial distribution of vegetation across the study area and compare the patterns produced by the two indices.

Objective

To calculate and visualize NDVI and EVI for Ibarapa East LGA and examine the spatial patterns of vegetation represented by the two indices.

Data Used

- Nigeria Local Government Area administrative boundaries - Vector layer
- Raster satellite imagery containing the spectral bands required for NDVI and EVI calculation: Two 10m Resolution Sentinel Data which are Near Infrared (NIR) i.e Band 8 and Red Band i.e Band 4.
- Area of interest: Ibarapa East LGA, Oyo State, Nigeria

Vegetation Index Formulas

Normalized Difference Vegetation Index (NDVI)

NDVI was calculated using the standard relationship between the Near-Infrared (NIR) and Red bands:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

Enhanced Vegetation Index (EVI)

EVI was calculated using the standard enhanced vegetation index equation:

$$\text{EVI} = 2.5 \times (\text{NIR} - \text{Red}) / (\text{NIR} + 6 \times \text{Red} - 7.5 \times \text{Blue} + 1)$$

The required spectral bands were used as raster inputs in QGIS.

Methodology

1. Imported the Nigeria Local Government Area administrative boundary dataset into QGIS.
2. Identified and selected Ibarapa East LGA in Oyo State as the study area.
3. Imported the required raster satellite data into QGIS.
4. Prepared the relevant spectral bands for analysis.
5. Calculated NDVI in QGIS using the Raster Calculator.

6. Calculated EVI in QGIS using the Raster Calculator and the required NIR, Red, and Blue bands.
7. Visualized the resulting NDVI and EVI raster layers using appropriate graduated raster symbology.
8. Compared the spatial patterns represented by the NDVI and EVI outputs.
9. Prepared the final map layout with relevant cartographic elements.

Analysis

The analysis provides a spatial representation of vegetation characteristics within Ibarapa East LGA. The NDVI and EVI outputs showed broadly similar spatial vegetation patterns but differed in the intensity and distribution of the vegetation signals. NDVI provided clearer differentiation between vegetated and non-vegetated surfaces, while EVI produced more continuous representation of vegetation variability across Ibarapa East LGA.

Key Outputs

- NDVI raster map of Ibarapa East LGA
- EVI raster map of Ibarapa East LGA
- Final cartographic print layout for the vegetation index analysis

QGIS Techniques

- Raster data processing
- Raster Calculator
- Vegetation index calculation
- Vegetation indices generation
- NDVI analysis
- EVI analysis
- Raster visualization
- Vegetation Map layout design

Skills Demonstrated

Remote Sensing | Raster Analysis | NDVI | EVI | Satellite Image Analysis | Cartography

Project Significance

The workflow demonstrates how satellite imagery and vegetation indices can be used to support environmental monitoring, vegetation assessment, land-cover studies, and spatial analysis.